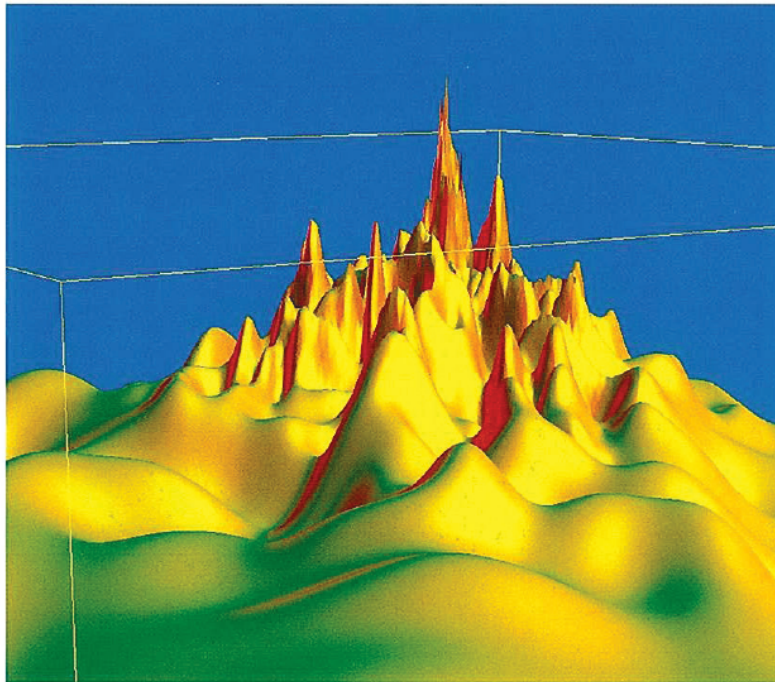


THIRD EDITION

STRATEGY
and the
BUSINESS
LANDSCAPE

PANKAJ GHEMAWAT



Editorial Director: Sally Yagan
Editor in Chief: Eric Svendsen
Acquisitions Editor: Kim Norbuta
Editorial Project Manager: Claudia Fernandes
Director of Marketing: Patrice Lumumba Jones
Senior Marketing Assistant: Ian Gold
Project Manager: Renata Butera
Creative Art Director: Jayne Conte
Manager, Rights and Permissions: Charles Morris
Cover Art: Andrei Linde and Dimitri Linde
"Computer simulations of stochastic processes

during the stage of cosmic inflation soon after the Big Bang."

Full-Service Project Management: Saraswathi Muralidhar, GGS Higher Education Resources
Composition: GGS Higher Education Resources, A Division of PreMedia Global, Inc.
Printer/Binder: Edwards Brothers Malloy
Cover Printer: Lehigh-Phoenix Color/Hagerstown
Text Font: 10/12 Palatino

Credits and acknowledgments borrowed from other sources and reproduced, with permission, in this textbook appear on appropriate page within text.

Copyright © 2010, 2006, 2001, 1999 Pearson Education, Inc., publishing as Prentice Hall, One Lake Street, Upper Saddle River, New Jersey 07458. All rights reserved. Manufactured in the United States of America. This publication is protected by Copyright, and permission should be obtained from the publisher prior to any prohibited reproduction, storage in a retrieval system, or transmission in any form or by any means, electronic, mechanical, photocopying, recording, or likewise. To obtain permission(s) to use material from this work, please submit a written request to Pearson Education, Inc., Permissions Department, One Lake Street, Upper Saddle River, New Jersey 07458.

Many of the designations by manufacturers and seller to distinguish their products are claimed as trademarks. Where those designations appear in this book, and the publisher was aware of a trademark claim, the designations have been printed in initial caps or all caps.

Library of Congress Cataloging-in-Publication Data

Ghemawat, Pankaj.

Strategy and the business landscape / Pankaj Ghemawat. — 3rd ed.

p. cm.

ISBN-13: 978-0-13-601555-0

ISBN-10: 0-13-601555-7

1. Strategic planning. 2. Industrial management. 3. Competition. I. Title.

HD30.28.G484 2010

658.4'012—dc22

2009019672

10 9 8 7



Prentice Hall
is an imprint of



**This book is not for sale or
distribution in the U.S.A. or Canada**

www.pearsonhighered.com

ISBN 10: 0-13-601555-7
ISBN 13: 978-0-13-601555-0

CHAPTER

5

Sustaining Superior Performance

But many that are first shall be last; and the last first.

—MARK 10:31

This chapter, like Chapter 4, focuses on dynamics. It takes a different track, however, which can best be illustrated by expanding upon the advice offered at the beginning of Chapter 3, to build the better mousetrap. We need to take that advice a step further and look at **sustainability in the face of imitation**. Because garages seem to feature many small mousetrap manufacturers as well as mice, an innovator will gain little from anticipating its interactions with any one rival in the kind of dynamic detail discussed in Chapter 4. But it *can* add a useful dynamic dimension to the analysis by recognizing that better mousetraps that have succeeded in the marketplace in the past (e.g., repeating traps and glueboards) haven't greatly enriched their respective innovators, partly because better mousetraps attract imitators as well as mice.

Recognizing the threat of imitation is the first step in thinking through whether barriers to imitation exist or can be erected around the innovation. And even if imitation cannot be blocked, its likely scope and speed should be accounted for in deciding whether to develop the better mousetrap in the first place. **This chapter discusses imitation and other generic threats to the sustainability of superior performance as well as strategic responses to those threats.**

Data on performance dynamics help illustrate the importance of thinking through issues of sustainability. Historically, such dynamics were understudied because they did not form part of the traditional strategy agenda laid out in the profitability grid in Exhibit 1.7. But in the last 15 years, evidence has accumulated

that points to the folly of focusing on achieving superior performance while forgetting about its sustainability. This evidence suggests that above-average performance tends to subside toward the average *much more rapidly* than many managers assume.

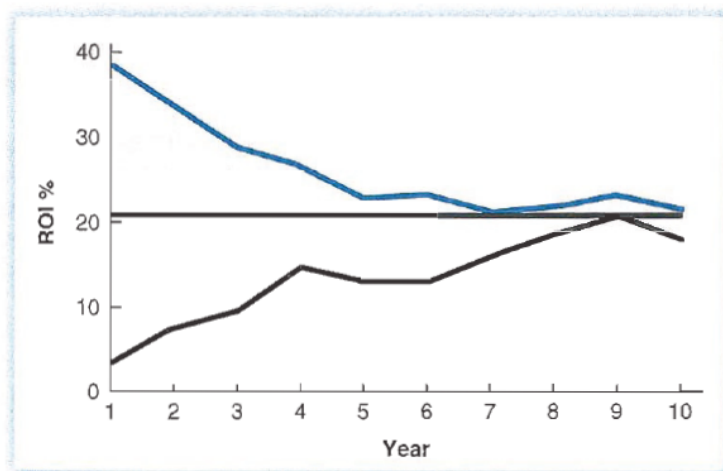
Consider, for example, an analysis of the return on investment (ROI) reported over a 10-year period by a sample of nearly 700 business units.¹ When this sample was divided into two equally sized groups based on initial ROI, the top group's ROI in year 1 was 39 percent and the bottom group's was 3 percent. If businesses were kept in the groups in which they started out and tracked over a decade, what changes would you predict to the 36-point ROI spread between the group averages by year 10?

When managers are confronted with this question, they often guess that the initial ROI spread between the two groups shrank by between one-third and two-thirds by year 10. In fact, as shown in Exhibit 5.1, the shrinkage was actually greater than 90 percent! In another recent study, only 2 percent of the firms in a large sample managed to sustain superior performance over at least a 10-year period.² Such data clearly imply that in most cases, the sustainability of superior performance cannot be taken for granted.

But one should not conclude that just because sustainability is elusive, it is not worth examining. There are at least four reasons why. First, even if initially superior returns can be expected to subside to the average level by year 10, the degree to which they can be sustained in the interim—the shaded area in Exhibit 5.1—is surely of interest. Second, convergence actually *was not* complete by the tenth year for the sample: the initially top group still had an ROI advantage of 3.5 percent. This is a significant difference in long-run spreads that can, in many situations, spell the difference between significant value creation and value destruction. Third, the convergence of returns overstates the convergence of what is actually of interest: (economic) returns times volume. A business posting an ROI of 39 percent—again, the initial average for the top group—probably should not insist that all new investments deliver returns

EXHIBIT 5.1 The Limits to Sustainability.

Source: Pankaj Ghemawat, *Commitment* (New York: Free Press, 1991).



that high. And finally, if you are still taken aback by the extent to which the group averages in Exhibit 5.1 converged, remember that they mask tremendous amounts of variation. It is easy to cite companies that have beaten the averages by large margins over much longer periods (e.g., Wal-Mart). Understanding how they have done so will help illustrate the range of possible strategic responses to threats to sustainability.

To that end, the next section classifies threats to the sustainability of superior performance and relates them to the analytical frameworks and perspectives developed in earlier chapters. Each of the four sections that follow focuses on (1) a threatening evolutionary dynamic and (2) possible strategic responses to it. The final section uses the example of Wal-Mart to review the discussion.

SUSTAINABILITY AND RESOURCES

In classifying threats to sustainability, it is useful to split observed performance into the two components suggested in Chapters 2 and 3, industry-level performance effects and within-industry performance differentials. Recent research suggests that superior performance that is sustained over the long haul is due, in large part, to stable industry-level effects, and that performance advantages over the competition tend to be less sustainable.³ Industry-level effects were discussed in some detail—from a dynamic as well as a static perspective—in Chapter 2 and therefore will not be taken up here, beyond the reminder that dynamics such as demand saturation, technological maturation, and resource depletion do need to be taken into account. The discussion in this chapter will focus, instead, on threats to the sustainability of superior within-industry performance and ways of dealing with them.

Research in industrial organization (IO) economics is helpful in this regard: it suggests that commitments to durable, specialized resources—what might be called sticky resources—are generally needed to sustain within-industry profit differences.⁴ The deregulation of the U.S. airline industry and its subsequent experiences illustrate this link. The U.S. airline industry was deregulated in 1978 in the belief that the threat of “hit-and-run” entry would keep prices and profitability in check even in city-pairs that ended up being monopolized. Individual city-pairs were supposed to be contestable, that is, easy to enter and exit, on the grounds that if prices exceeded relevant costs, an entrant could fly its capital into a market and, if threatened with retaliatory price cuts, could fly it out again. (As Alfred Kahn, who headed the Civil Aeronautics Board at the time of deregulation, put it, airplanes were “marginal costs with wings.”) But more than 25 years later, the U.S. airline industry still continues to exhibit significant, sustained within-industry differences in profitability. So the logic of contestability is revealed to be wrongheaded in the industry that was once supposed to be its leading exemplar. But the airline industry is worth examining, nonetheless, to see why differences in profitability have not disappeared.

The summary answer is that the contestability theory overlooked a range of sticky resources that have turned out to be important in the deregulated airline industry. Actually, two distinct phases can be discerned in the industry’s evolution since deregulation. Through the 1980s and part of the 1990s, the established airlines fended off entrants much more effectively than the deregulators had anticipated, largely by building and leveraging such resources as hub fortresses, computerized reservations systems, and customer stickiness based on frequent-flyer programs. But as prices have continued to fall—to less than half the real levels of the mid-1970s—the

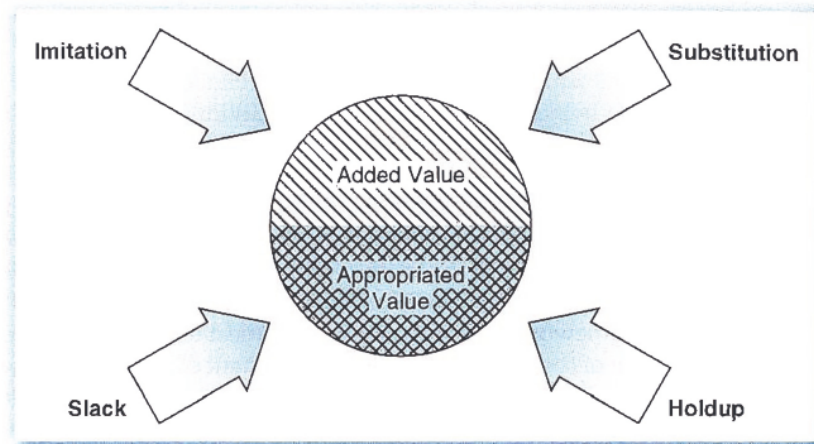
not-so-efficient resources that the incumbents continue to be stuck with, particularly expensive unionized workforces, have loomed ever larger and are, in several cases, the focus of their current struggles to survive. The discount carriers that have challenged the traditional incumbents can also be looked at through the resource lens. Southwest, by far the largest creator of economic value in the airline industry after deregulation, has a uniquely standardized and fuel-efficient fleet of planes and has historically managed to keep them flying one-third longer each day than its competitors. And even the pressures that it has recently experienced related to *its* employee costs draw attention, once again, to sticky resources.

This emphasis on resources has a long tradition in the field of strategy as well as in IO economics.⁵ A resource-based perspective does, however, depart from the emphasis in the previous chapters, particularly Chapter 3, on the activities that a business performs. To clarify, the present chapter shares with the others in this book an underlying stocks-and-flows conception of resources and activities. Specifically, resource stocks that can be varied only in the long run—in keeping with the traditional economic conception of resources as fixed inputs—determine the flows of activities that a business can undertake in the short run and these activities, in turn, help shape the evolution of resource stocks over time. Reinterpreted in these terms, Chapter 3 exploited the idea that even when the underlying resources are key, an activity lens may still be useful because it is hard to value fixed inputs except in terms of the flows of services they generate. In other words, activities are easier to measure and, therefore, often a better basis for analysis. But for explicitly dynamic purposes, such as understanding the determinants of sustainability, it is important to look directly at long-run factors. This involves shifting the focus from activities that can be varied in the short run to the underlying resource stocks that are fixed in the short run.

Resource-based research in strategy was slow to capitalize on these insights, partly because of an early tendency to define virtually anything as a resource.⁶ As a result, traditional resource-based frameworks for thinking about sustainability often lack a clear conceptual basis.⁷ Contemporary resource-based research advances matters by confirming the intuition that in thinking about the conditions under which resources will or will not continue to sustain superior performance, it is useful to distinguish between their added or scarcity value and their appropriability, or capture, of (some of) that value by the focal firm.⁸ Note that these two kinds of conditions—one related to competition and the other to bargaining—are meant to be overlaid: the sticky resources that underpin superior performance at a point in time must continue to be scarce to sustain superior performance, *and* the owners of the firm must be able to appropriate some of that sustainable scarcity or added value.

The discussion in the rest of this chapter will be organized in terms of two threats to scarcity or added value and two threats to appropriability. Specifically, it will work through the tetra-threat framework for diagnosing the sustainability of superior within-industry performance that is laid out in Exhibit 5.2:

- **Imitation** of the resources underpinning superior performance to the point where they are no longer scarce is a direct threat to the sustainability of added value. Imitation can be thought of as diminishing the extent to which the player being imitated would be missed if it simply disappeared—a rough measure of added value, as noted in Chapter 3.

EXHIBIT 5.2 The Tetra-Threat Framework for Sustainability Analysis.

- **Substitution** of the resources underpinning superior performance is an indirect threat to the sustainability of added value, driven by displacement rather than duplication. That said, substitution-related threats to scarcity are often even harder to address than threats of imitation.
- **Holdup** is a threat to the appropriation, or capture, of sustainable added value that is often rooted in resource cospecialization. Cospecialization can be essential to value creation, but it does create a problem: as a firm and its complementor (for instance) cospecialize, their added values overlap, making it impossible for both of them to appropriate the full amount. Each then faces the threat of being held up by the other.
- **Slack** is the internal threat to the appropriation, or value capture, that reflects a persistent tendency to dissipate potential economic surplus. The scope for slack depends on past success or current added value: firms without much of either have a hard time surviving while persistently suboptimizing.

Since the tetra-threat framework is superficially similar to the “five forces” framework (imitation ↔ rivalry, substitution, holdup ↔ supplier/buyer power), the differences are worth illustrating. Compare the application of the two frameworks to the pharmaceutical industry, which was analyzed with the help of the five forces framework in Chapter 2. A list of rivalry determinants would flag high levels of concentration, particularly within therapeutic categories, significant product differentiation, rapid growth, and so on as contributing to the pharmaceutical industry’s high average profitability. But looking at where the big money is made *within* the industry focuses attention on the blockbuster drugs that have come to dominate big pharmaceutical companies’ sales and profits. Why aren’t these imitated more effectively than they seem to be? Addressing this question requires attention to patents and other barriers to imitation, also rooted in underlying resources, which actually limit the number of sellers of such drugs.

Similarly, there is a big difference between the product-substitution threats which the “five forces” framework emphasizes and the broader resource substitution threats—ranging from entrants using new drug-discovery technologies to the open science movement—that are one of the components of the tetra-threat framework. Considerations of cospecialization and holdup highlight reliance on a range of suppliers of key resources: biotech and other focused firms, individual inventors, academia, and the National Institutes of Health, among others. Without such a focus on resources, five forces analyses of supplier power in pharmaceuticals tend to split attention across suppliers of variable as well as fixed inputs—or sometimes even ignore the latter entirely. And finally, slack has no counterpart in the “five forces” framework.

In addition to these differences in focus, the two frameworks also differ in terms of concreteness about the behavioral processes that threaten sustained superior performance. The “five forces” framework leans heavily on statistical tendencies and regularities (e.g., the concentration of competitors as a key determinant of profitability), whereas the behavioral foundations of the tetra-threat framework are clearer (e.g., the usual assumption that imitation of a scarce resource will occur if the effort can pay for itself).⁹ In other words, tetra-threat analysis operates at a level in between the macro/statistical perspective of the “five forces” framework and the microperspective, involving detailed, grounded analyses of individual players in specific situations, that was presented in Chapter 4. Given differences in the issues flagged, analysis at multiple levels is warranted, and hence this chapter should be seen as a supplement to the earlier ones.

THE THREAT OF IMITATION

Imitation is not always bad for an innovator. It can sometimes lend credibility to a new product (e.g., when network externalities or second-sourcing is an issue). In addition, the imitation of certain types of marketing moves, such as loyalty schemes and meet-the-competition clauses, does not necessarily undermine their effectiveness.¹⁰

Still, imitative development of scarce resources is usually bad news in added-value terms. And it seems to be endemic. Attempts by one player to increase its capacity, for example, often trigger additions by competitors intent on preserving their capacity shares. Attempts to build one’s customer base tend to prompt competitors to defend or develop their own. Furthermore, attempts at product differentiation based on R&D as opposed to marketing are vulnerable on several counts; studies indicate that competitors secure detailed information on most new products within one year of their development, patent-based strategies usually fail to deter imitation, and imitation tends, on average, to cost one-third less than innovation and to get to market one-third faster. And process innovations seem, if anything, to be even more imitable than product innovations.¹¹

The upshot is that the threat of imitation has to be taken seriously, even in contexts where the barriers to imitation might seem to be high. The pharmaceutical industry, for example, is characterized by product patents that are unusually effective at choking off imitation for periods exceeding a decade. But the big pharmaceutical firms still have to worry about imitation of the limited number of blockbuster drugs—typically less than five—on which their fortunes depend. (This dependence was dramatized by Merck’s 2004 recall of Vioxx, its fourth-best seller in 2003 with \$2.5 billion in revenues, triggering a \$27 billion, or 27 percent, drop in Merck’s market value the day the recall was announced.)

Some imitative pressures reflect the big pharmaceutical companies' practice of tweaking each other's innovative blockbusters to come up with slightly different molecules that are not protected by the original patents, which they then patent and sell under their own brands. But the effects of such copycat molecules are moderated by the branded manufacturers' ability to contain price competition among themselves, perhaps because they recognize their mutual dependence, as in the pharmaceutical pricing game described at the beginning of Chapter 4. What are much more threatening to high prices are the generic manufacturers who enter with 30–90 percent discounts after patents expire, and typically manage to wrest the bulk of category volume away from the original patent holder in a matter of months.¹² Concerns about generic competition loom particularly large among branded manufacturers as of this writing due to impending patent expirations and an anticipated dearth of new blockbusters over the next few years.

The diverse ways in which branded pharmaceutical companies have tried to mitigate the threat of imitation illustrate a variety of barriers to imitation, often involving early-mover advantages, that extend well beyond a prespecified period of patent protection and are at least partly endogenous, meaning that they can be influenced. **The following barriers should not be seen as mutually exclusive: strong defenses against imitation typically rely on more than one of them.**

Private Information

To the extent that **superior information or knowledge can be kept private—to the extent that it is costly for would-be imitators to tap into it—imitation will be inhibited.** Although being tight-lipped can sometimes ensure privacy, many other channels of potential informational leakage exist, including customers, employees, and other suppliers, spinoffs, reverse-engineering, and even patent documents. As a result, privacy is most likely to be sustainable when information is tacit rather than specifiable (i.e., doesn't lend itself to blueprinting), and when no one party can carry it out of the organization. Thus, Wyeth has managed to monopolize a lucrative hormone-replacement submarket with its Premarin drug, made from the urine of pregnant mares, for more than 60 years now even though the drug's patent expired decades ago. A two-decade-long attempt to develop a synthetic generic version of the drug has been thwarted by the fact that nobody knows exactly what is *in* Premarin.¹³

Switching Costs/Relationships

The severity of imitation threats may be blunted by **signing enforceable contracts or, more commonly, creating switching costs or cultivating relationships that secure superior access to customers, suppliers, or other important types of players.** The branded pharmaceutical companies have been particularly imaginative in engaging doctors, for example.¹⁴ They induce trials by doctors (and hospitals) of their newest and most expensive products by distributing billions of dollars of free samples. In addition, they give doctors personal gifts, provide them with generous "research" grants, and offer them consulting, speaking, and even publishing opportunities. This largesse can be thought of as leaving a residue of goodwill with doctors, with effects that linger on in their prescription decisions even after generics become available.

Size Economies

Size economies refer to the (possible) advantages of being large. They come in at least three different varieties: **scale economies**, which are the advantages of being large in a particular business at a particular point in time; **economies associated with learning by doing**, which imply advantages to being large in a particular business over time; and **scope economies**, which are the advantages of being large across inter-related businesses and which are discussed in more detail in Chapter 6.¹⁵ If there *are* size economies, a firm may be able to deter imitation by precommitting itself to exploiting them. Deterrence depends on the possibility that if the would-be imitator tried to match the incumbent's size or resource mass, enough excess capacity might result to make it regret the effort. The pharmaceutical majors' sales forces stand out in this context: there are indications that would-be entrants shy away from categories in which they would have to achieve similarly broad reach (typically ones in which drugs have to be promoted to legions of primary-care physicians). Scale economies are also evident in the shift from print to TV advertising of prescription drugs after the rules on direct-to-consumer TV ads were relaxed in 1997.

Threats of Retaliation

Incumbents may also be able to deter imitation by threatening massive **retaliation**. But talk of retaliation is cheap. To be credible, it must be backed up by both the ability and the willingness to retaliate. **The ability to retaliate is facilitated by the creation of a competitive advantage and by the maintenance of resource reserves, such as liquidity, excess capacity, "warehoused" product upgrades, and positions in competitors' other businesses, that can be used to put pressure on them. The credibility of retaliation is greatest** when it is payoff-maximizing for the retaliator, but it *can* be boosted in several ways: by writing contracts that make retaliation more attractive than accommodation, otherwise binding oneself (e.g., "burning one's bridges"), cultivating a reputation for retaliating in spite of immediate losses to oneself, and so on. Thus, branded pharmaceutical companies threatened with patent expiry now often license their drugs to makers of "authorized generics" or sell their own generics through subsidiaries so as to disrupt the six months of exclusivity that are granted to the first generic manufacturer under U.S. law and are often critical to the pioneering generic manufacturer recouping its investments.¹⁶

Imitation Lags

Even if the barriers to imitation discussed earlier do not apply, imitation typically takes time. Guesstimates of the average lead times in taking various types of actions will help underscore the importance of such lags.¹⁷ As a rule of thumb, prices and marketing communications are among the few variables that can be changed significantly in less than one year. Changes to resource stocks tend to take longer. It takes two years or more to build and start up the typical manufacturing plant. Building a new distribution system or altering an existing one seems at least as time-consuming. Studies indicate that the mean lag in returns from R&D tends to be on the order of four to six years. And the lags in implementing major changes in human resource practices and restructuring corporate portfolios may stretch out over the better part of a decade—or longer! Changes along multiple dimensions are likely to take longer (and have a higher probability of failure), especially if they have to be executed in sequence.

In pharmaceuticals, the key lags concern generic entry, with each day of delay potentially being worth millions of dollars to the owner of a blockbuster. Instead of treating their time-in-market before generics enter as given, the pharmaceutical majors have taken a range of actions to lengthen it: “evergreening” their patents by tweaking molecules, uses, or formulations, suing generic manufacturers for patent infringement (which automatically delays approvals of generic drugs in the United States by thirty months), exploiting other loopholes that extend patent protection, such as testing drugs on children, which is worth six months, lobbying for the general lengthening of pharmaceutical patents, pressing the Food and Drug Administration to speed up the drug-approval process, and so on. The results? It has been estimated that the effective time-in-market for brand-name drugs before generic entry increased from about eight years in 1980 to 14 years in 2000.¹⁸

Upgrading

Another significant barrier to imitation is based on **continuous upgrading of the organization’s own added value**. Operationally, this involves **driving a wider wedge between customers’ willingness to pay and suppliers’ opportunity costs** over time, and often requires significant resource commitments. Thus, the pharmaceutical majors continue to make major commitments to developing and introducing new blockbusters at costs that are now estimated at \$1.7 billion per successful drug and have also been paying more attention in recent years to upgrading their product and process development capabilities (see Exhibit 5.3 for some additional discussion of commitments and capabilities). **Note that upgrading makes a business a moving target in a way that compounds the difficulties or delays for potential imitators and may even permit the upgrader to break away from the pack.** One way to calibrate the need to upgrade is to track the rate at which an industry’s real prices, adjusted for quality, change over time. If an industry’s average prices decrease by more than a threshold rate (3 percent per year, according to Jeffrey Williams), it is a “fast-cycle” environment in which advantages tend to be short-lived rather than sustainable, and more or less continuous upgrading is required.¹⁹ The bias toward action built into upgrading also reminds us that barriers to imitation are not just forms of protection that lucky firms are blessed with; they *can* be built.

Numerous other types of barriers to imitation have also been proposed. Several are rooted in considerations of complexity, including causal ambiguity,²⁰ as evoked by Wyeth’s Premarin, social complexity²¹ that is supposed to put certain resources—corporate culture is the favorite example—beyond the reach of systematic management, and the challenges of achieving cross-sectional fit, or coherence, across many different types of choices.²² But while insurmountable complexities can help explain barriers to imitation *ex post*, their implications for action *ex ante* are often unclear. For instance, even if it is more difficult for an imitator to copy a whole set of coherent choices instead of just one or two, is it obvious that the innovator had an easier time coming up with the coherent set in the first place? If not, there is no point to pursuing sustainability (as opposed to advantage) through intricate alignment of choices because there are no early-mover advantages to be gained thereby.

In summary, imitation is a direct threat to the sustainability of added value whose severity is inversely related to the height of barriers to imitation. Such barriers often revolve around early-mover advantages. But while moving early may reduce the threat of imitation, it can also raise costs and risks. As a result, the only sensible

EXHIBIT 5.3 Making Commitments and Building Capabilities.

The resource-based view of strategy takes a historical perspective by stressing that the resources a business has assembled determine its menu of opportunities at any point in time. In this sense, it links what the business did in the past to what it can do well today. But to stop there would be to confine attention to the exploitation of “legacy” resources—which would be seriously incomplete, since the value of the resources that firms already have in place is often estimated to account for less than one-half or even one-quarter of their market value.²³ Thus, in pharmaceuticals, while it is extremely important for firms to exploit their legacies—blockbusters still under patent—for all they are worth, it is also essential to invest for the future by making large, relatively discrete **commitments** of resources to developing, testing, and marketing new drugs as well as more diffuse commitments to improving their R&D and manufacturing capabilities, among others.

By implication, a fully dynamic approach to strategy must not only link what a business did in the past to what it can do well today but also what it does today to what it can do well in the future. This involves explicitly considering both lumpy commitments and **capabilities** as ways for it to upgrade or shift its opportunity set. Lumpy commitments are a good way of thinking about not only market entry (and exit), major new-product introductions, significant capacity expansion/modernization, and so on, but also some less obvious categories of change, such as cultural transformation and even shifts in strategy. Thus, when Coors switched from a regional to a national strategy in the U.S. beer market in the 1970s, it had to open its first new manufacturing facility in a century, establish relationships with distributors in 39 new states, improve acrimonious relationships with unions and minorities that were stronger in the new states, build a national clientele by spending more per barrel on advertising than entrenched competitors (two of which, Anheuser-Busch and Miller, also benefited from greater national scale), upgrade its marketing capabilities, and change a production-focused culture. The Coors case also illustrates the four markers of irreversibility, the true measure of significance of lumpy commitments:

- sunk costs that would be unrecoverable if going national did not work out
- opportunity costs that induce lock-out instead of lock-in, as illustrated by Coors locking itself out of certain opportunities by moving very late to go national
- time lags of longer than a decade in completing the switch to a national position
- organizational commitment to or momentum behind the new course of action that would make it hard to reverse even if reversal turned out to make economic sense

Lumpy commitments are individually important enough to warrant the full power of the analytical techniques discussed in this book—assessing fit with or influence on industry structure, and implications for competitive position, interactions, and sustainability.²⁴

The development of capabilities, in contrast, involves choices that are individually small and frequent rather than individually important and infrequent. The idea is that firm-specific opportunities to perform activities more effectively than competitors or to make commitments on better terms than they can be built gradually and reinforced over long periods of time.²⁵

For a vivid example, consider the success of Japanese automakers in the U.S. market. It is hard to explain this success in terms of superior product market positioning: sedans like the Honda Accord and the Toyota Camry were priced and targeted similarly to the domestic automakers’ core offerings. More explanatory power is provided by the estimate that by the 1980s, Japanese automakers had upgraded their product-development capabilities to the point where they could design and introduce a

(continued)

EXHIBIT 5.3 Continued

new model in about *half the time* and with *half the engineering resources* required by their U.S. counterparts. This allowed them to update their products more frequently. They, especially Toyota, also managed to produce their products more cheaply and with better quality because of differences in manufacturing facilities and processes. The still unmatched Toyota Production System is protected by knowledge that is tacit and rooted in complex organizational processes, economies of plant scale (particularly in Japan), relationships with suppliers that are elaborated on in the section on holdup, time lags for imitators to build up comparable resources, and a deep commitment to continuous improvement.

Many different kinds of capabilities can be identified, associated with differences in product and process development efficiency, learning, speed, flexibility, mobility (across lines of business), and so forth.²⁶ "Metacapabilities" associated with integration, reconfiguration, and transformation have also attracted some interest but are quite hard to measure. But measurement and comparisons with competitors are important because in their absence, hubris and politics are likely to lead to excessively high self-ratings and a tendency to designate anything one cares about as a key organizational capability.²⁷

Another challenge relates to the incremental nature of capability development: firms must prevent the overall coherence of such efforts from being nibbled away, choice by choice, by drop-in-the-bucket biases, and the like. Somewhat paradoxically, this makes the choice of which capabilities to develop, and how, a relatively lumpy choice, just like conventional resource commitments, since commitment to a major capability-development thrust also exhibits lock-in, lock-out, lags, and organizational commitment. As a result, just like specific commitments, specific capabilities also imply specific rigidities that can, in an uncertain world, result in inferior rather than superior performance.²⁸ In other words, capability development and commitment may be necessary for sustained success, but are certainly not sufficient. Which is why they must be analyzed.

conclusion about early versus late timing is the one drawn long ago by Alfred P. Sloan: If you are late, you have to be better.

THE THREAT OF SUBSTITUTION

Added value may be threatened by substitution as well as by imitation. **Substitution should be thought of here in terms of resource substitution, rather than the product substitution highlighted in Chapter 2. Resource substitution tends to reduce added value by displacing scarce resources instead of duplicating them,** that is, by reducing the demand for their services instead of expanding their supply. And while substitution can sometimes blur into imitation at the margin, it can also have rather more drastic effects, as emphasized more than half a century ago by the economist Joseph Schumpeter, who compared the threats of substitution and imitation "as a bombardment is in comparison with forcing a door."²⁹

Substitution seems to command much more attention among business strategists today than it did in Schumpeter's era. Particularly popular are discussions of **disruptive technologies**.³⁰ Substitution is a related but broader idea. Thus, technological substitution encompasses not only disruptive technologies that, while suboptimal for the requirements of existing customers, are improving more rapidly, but also technological threats from above (as opposed to below). Thus, in the first 20 years of the videogame market, each new generation of consoles led to a new

competitor becoming the market leader. Of course, the even more radical change associated with Nintendo's wireless handheld controller and dual-screen system *could* be seen as a threat from below, underlining the point that substitution threats can come from both directions. Furthermore, as a threat to added value, substitution encompasses much more than just technological change. Other "supply-side" triggers of substitution include changes in input prices or availability and deregulation (as in the airline industry). On the "demand side," shifts in customer preferences, previously unmet needs, and changes in the customer mix can shake up things.

i For a detailed example of a threat of substitution and responses to it, consider the growth of online book retailing in general and Amazon in particular from the perspective of the leader in traditional (bricks-and-mortar) book retailing, Barnes & Noble. Amazon's online store threatened to largely dispense with and displace two of the key resources that Barnes & Noble was stuck with: approximately 10 million square feet of physical selling space in stores whose value, properly capitalized, exceeded the company's annual revenues, and 15,000 in-store employees (1998). What Barnes & Noble did and did not do illustrates the broad array of possible responses to substitution threats: acquiesce, defend, straddle, switch, and innovate.

Not Responding

One way of dealing with substitution dynamics is by not responding at all. This is *not* what Barnes & Noble did: within a year of Amazon's founding, it decided to launch its own online operation. Still, one can imagine some plausible arguments in favor of inaction:

- Online book retailing may prove to be just a flash in the pan
- Even otherwise, online will amount to only a small niche
- The online model underserves purchasers because it does not let them handle books or provide the physical ambiance emphasized by Barnes & Noble's superstores
- Online will cannibalize the physical stores, which are already under pressure
- Barnes & Noble does not have the skills and expertise to make an online initiative work

These objections typify the arguments that are used more broadly to limit consideration of substitution threats.³¹ But the general sense—bolstered by recognition of such corporate sins as arrogance, myopia, bureaucracy, and sloth—is that paying too little attention to substitution threats is often more hazardous to a company's health than paying too much.

Migrating/Harvesting

Migrating and harvesting, often employed in tandem, are less passive forms of acquiescence. **Migration involves redeploying resources to uses that are less vulnerable to substitution threats or more attractive overall in light of them. Harvesting requires a shift toward milking existing resources instead of building or even maintaining them.** Research into disruptive technologies suggests that while harvesting is the most common response of incumbent firms, it is often accidental, at least in the early stages of the process.³² On the other hand, it is also easy to imagine cases in which harvesting would have made sense but was not attempted. For example,

questions can be raised about whether Barnes & Noble should have continued to invest in expanding its network of physical stores despite persistently poor returns.

Defending

Defending against substitution threats means that **the threatened business must either increase its willingness to pay or decrease its costs**. The obvious danger with this approach, especially in the context of technological threats, is that the threats may have fundamentally faster improvement dynamics.³³ An implication from Chapter 3 that may be less obvious is also worth emphasizing: defense can be achieved not only by improving one's own economic fundamentals but also by impairing the fundamentals of competitors. Thus, Barnes & Noble attempted to squeeze competitors, particularly Amazon, in a number of ways, most notably by announcing, in November 1998, that it would acquire Ingram, which accounted for more than half of book wholesalers' sales in the United States and was also a major supplier to Amazon and others. This deal had to be called off, however, after complaints from publishers and book retailers prompted the U.S. Federal Trade Commission to launch an investigation.

Straddling

Straddling **involves responding to a substitution threat by establishing a foothold in both camps**. Straddling may be either a transitional hedge or a long-term strategy. Straddles can also be distinguished in terms of whether they are relatively balanced between the old and the new or are imbalanced between the two (e.g., offer only a toehold in the substitute form of competition). Perhaps the biggest risks inherent in straddling are excessive commitment to an old form of competition that may no longer be viable and an unwillingness to make tough choices. Straddling seemed to be Barnes & Noble's principal strategy between 1996 and 2000: it moved to set up Bandn.com as a leading online retailer (rather than a toehold) by drawing on talent and content resources from outside the company, but kept the online venture separate, organizationally and operationally, from its traditional operations.

Switching

Switching is a more radical response to substitution threats than straddling because it involves **a wholesale shift to the substitute**. Despite its obvious attraction in terms of speed—and the enthusiasm for it in the disruptive-technology literature—switching is, especially in fast-moving environments, like changing horses in mid-stream: there is sometimes no alternative, but the maneuver is often subject to spills. Switches also raise questions that should be asked and answered *before* the fact about how the business that is switching will add value after doing so. Given its sense that online retailing would continue to be a niche within the broader book market, it is not surprising that Barnes & Noble chose not to switch.

Recombining

Switching involves adoption of a model of how to compete that is already “out there.” In many cases, however, **recombining elements of existing ways of competing with some of the new possibilities implicit in substitution threats seems to work better than wholesale switching**. Recombination or hybridization possibilities tremendously expand the range of possible responses to substitution threats.

Barnes & Noble should probably have paid more attention earlier on to such possibilities, particularly to the obvious ones of combining its unmatched store network with elements of online book retailing by, for example, using the stores to facilitate orders, delivery, and returns for the online business. The company finally began to move in this direction in late 2000 by introducing Internet Service Counters in the stores and allowing Bandn.com products to be returned there—but this move came too late to allow it to challenge Amazon for leadership online.

Leapfrogging

Leapfrogging is the most radically innovative response to a threat of substitution: it involves trying to outsubstitute the substitution threat by looking for a performance improvement or value innovation that promises even better performance.³⁴ Thus in 1999/2000, as it became clear that Bandn.com's Amazon-like approach was not gaining ground on Amazon, Barnes & Noble invested more than half a billion dollars in two attempts at leapfrogging: an e-book initiative that required expensive software but could potentially displace physical distribution and a print-on-demand initiative that required expensive printing machinery but could potentially displace some of publishers' traditional roles. Given their desperate character, it is not very surprising that neither initiative panned out.

In summary, there are no foolproof responses to substitution threats. Nevertheless, recognition of the variety and seriousness of such threats and the array of possible responses to them should help companies go beyond the fight-or-switch dichotomy.

THE THREAT OF HOLDUP

Even if an organization can protect its added value from the threats of imitation and substitution, the ability of its owners to appropriate that added value cannot be taken for granted. There are two systematic threats to value appropriation over time: holdup and slack. Holdup, discussed in this section, threatens to divert sustained added value to buyers, suppliers, complementors, and the like, whereas slack—discussed in the next section—threatens to dissipate it.

For a simple example of holdup, consider the privately owned Ambassador Bridge, which links Detroit in the United States and Windsor in Canada and handles nearly 30 percent of the annual trade between the two countries, which are the world's largest trading partners. Since the nearest competing bridge is 80 miles away, and farther from the region's economic center, the Ambassador can charge a significant premium: nearly twice as much for a gas tanker, according to a comparison from 2003.³⁵ Given the significant escalation in fuel prices since then, the Ambassador Bridge's locational advantage has expanded rather than eroded.

The scarcity value of the Ambassador Bridge is threatened by plans to build a second bridge close by to relieve congestion caused by the tightening of borders after September 11, 2001. But while plans for this new bridge are still being finalized, the owners of the Ambassador Bridge have invested significantly in amenities such as more customs booths and better highway connections, and have secured some of the permissions required to add a second six-lane span to the existing bridge—all the while projecting that total traffic will remain flat over the next 40 years.³⁶

The Ambassador Bridge's ability to hold up and appropriate some of the value created through cross-border trade is easy to analyze because the size of the pie is relatively fixed and the balance of power in determining how it is sliced is very one-sided. Holdup assumes more complex forms when players' actions affect the size of the pie as well as the way in which it is sliced up and when they exhibit interdependence, often reflecting underlying resource cospecialization.³⁷ Bilateral monopoly represents the extreme case in which the added value of each player is equal to the total value that the two create by transacting with one another.

To reflect on these complexities, consider the relationships between automakers and their parts suppliers. Currently, efficiency requires substantial tangible and intangible investments by both automakers and their suppliers, with underinvestment potentially raising costs and reducing differentiation by limiting quality and innovation, lengthening development cycles, and even raising recall risks. The required investments are often relationship specific and hard to reverse or recoup. But a supplier that modifies its designs, machinery, or processes to cater to a specific automaker's requirements runs some risk of being held up by it. Conversely, an automaker can leave itself vulnerable to holdup by its suppliers. Fears of holdup can, as a result, shrink the total size of the pie by reducing the amount of investment or by distorting it (by, for example, forcing inefficient reductions in investment specificity).

The evolution of automaker-supplier relationships over the last 100 years also illustrates a range of approaches for dealing with complex holdup threats.

Contracting

At the beginning of the twentieth century, U.S. automakers relied on older, more established suppliers to the bicycle and carriage industries.³⁸ As parts suppliers became more committed to the automobile industry and automakers undertook more coordination, the two sides shifted from transaction-by-transaction interactions toward longer-term relationships governed by contracts. Specialization subjected market-based, arm's-length transactions to market failures of the sort discussed in Chapter 6 and mandated alternative governance mechanisms. Contracting was one solution, but only a partial one, since completely comprehensive contracts enforceable at zero cost—contracts of the sort that could entirely eliminate fears of holdup—are typically impractical.³⁹

Integrating

A second response to holdup threats, integration, came to the fore with the rapid expansion and consolidation of the U.S. automobile industry. Total production more than quadrupled between 1909 and 1914, and Ford achieved market dominance by increasing its volume more than 25-fold with its inexpensive Model T. Ford also rethought its supplier relationships over this period. In 1909, its operations involved mounting a body, wheels, and tires (sourced from others) on a complete chassis supplied by the Dodge brothers. But by 1914, the Dodges were using their profits from the Model T to enter auto assembly, while Ford was vertically integrating backwards into parts and inputs, partly to reduce dependence on a few suppliers but also as part of a broader cost-leadership strategy. Vertical integration was meant to help efficiency by ensuring large-scale investments that suppliers were reluctant to make, easing coordination, exploiting

the potential for speed unlocked by Ford's assembly line innovation, and reducing the risk of disruption (no small concern, since Ford's operations were often within a half-hour of suspension because of tardy deliveries). By the mid-1920s, Ford's vertical scope extended all the way back into mines, plantations, forests, and even sheep farms.

Vertical integration combined with specialization, scale, and process innovation to help reduce the real costs of the Model T by three-quarters between 1909 and 1924, as illustrated in Exhibit 1.3. But it also helped triple asset-intensity, increased operating leverage, and—once heavier closed bodies began to replace the Model T's open-car design in the 1920s—proved very inflexible: Ford lost long-term leadership to General Motors (GM) as it changed over from the Model T to the Model A in 1927–1928.⁴⁰ Other vertically integrated auto operations also experienced pressure from independent suppliers' excess capacity and product innovation capabilities in the 1920s and 1930s. Furthermore, vertical integration did not so much solve coordination or incentive problems as transfer them from outside to inside the firm (e.g., the thorny issue of transfer pricing). In addition, vertical integration to avoid holdup by parts suppliers increased exposure to holdup by the industry's formidable labor union, the United Automobile Workers (UAW).

Increasing (and Using) Bargaining Power

After World War II, despite some deintegration at Ford, overall levels of integration remained high. GM, Ford, and, to a lesser extent, Chrysler manufactured at least some of their requirements of every basic component (and up to 100 percent in the case of such critical items as engines, transmissions, and axles), while outsourcing the rest to a number of suppliers with whom they played hardball.⁴¹ The automakers fragmented their supplier base and further reduced suppliers' added values by maintaining large in-house R&D staffs, forcing innovators to part with their technologies, breaking down systems into parts, and making every part into a commodity through comprehensive specifications. Contracts focused on prices, rarely ran for more than one year, and often were not renewed. Other ways in which the domestic Big Three put pressure on prices included using inspection teams to estimate suppliers' costs, letting small, low-overhead suppliers make low bids that could be used in negotiations with other suppliers, linking price concessions on a particular part sourced from a large supplier to continued purchases of other parts, and even starting rumors about potential competition.

Most of these practices represented attempts to counter supplier power, as discussed in the context of the "five forces" framework, so that framework does have something useful to say about the threat of holdup. Note, however, that taking just this bargaining/hardball perspective raises the risk of going too far in marginalizing suppliers, to a point where the shrinkage in the size of the pie outweighs the increased share of the pie that such efforts might secure. The other approaches discussed in this section remind us that maximizing relative bargaining power is not the only way to address the holdup problem.

Building Mutual Dependence

Large-scale entry by Japanese automakers into the U.S. market in the 1970s and 1980s forced the domestic Big Three to rethink their hardball approach, especially as the role of supplier relationships in the success of the Japanese became clearer.⁴² Despite less vertical integration than their U.S. counterparts, Japanese automakers worked

with far fewer suppliers with whom they maintained long-term relationships, exchanged employees and information, and partnered in innovation as well as production. Specifically, most parts supplied were “black boxes,” with the automaker providing only general specifications and the supplier preparing all of the detailed specifications and blueprints, which made it difficult to switch suppliers. The suppliers, in turn, invested in assets specific to individual automakers: the typical Japanese supplier-partner relied on a single automaker for 60 percent of its sales, versus a figure of 34 percent in the United States.⁴³ In other words, both sides built up their confidence in each other to the level where they were both willing to invest in a profit stream that would materialize only *if* they continued to work together. Such **interorganizational relationships** can be self-sustaining only if each player is allotted a sufficiently large share of the gains from cooperation. Opportunism may then be limited by the threat of disappearance of the large profit stream available from cooperation.⁴⁴ This is very different from the approaches discussed so far in this section, all of which emphasize the minimization of dependence on others.

During the 1990s, the domestic Big Three moved to cooperate more with their suppliers. GM and Ford spun off their large internal parts operations, Delphi and Visteon, helping address independent suppliers’ fears that any improvements they proposed would be passed on to the internal parts operations and their value thereby held up. But Chrysler, the least integrated to begin with, moved farthest in reshaping its supplier relationships by investing in coordination, recognizing suppliers’ needs to make a fair profit, soliciting feedback and sharing the savings, shifting toward longer-term contracts, and creating the expectation of business beyond the life of the contract if suppliers performed. GM and Ford never got as far and, in the grip of the industry downturn of the early 2000s, all of the Big Three began to revert to focusing mainly on prices.⁴⁵ Since then, the U.S. auto parts industry has experienced a wave of bankruptcies (including Delphi’s), divestitures, and restructuring—and refocused on other automakers whose output is actually growing.

Developing Trust

The stability of cooperative relationships can also be enhanced by trust. The Japanese business environment, for example, seems to deal with opportunism by placing more emphasis on social institutions (norms, etc.) and less on legal institutions than in the United States. It is possible that Japanese automakers have benefited from this general social context in establishing cooperative relationships with their suppliers. However, researchers continue to disagree about whether the Japanese auto industry *really* exhibits trust in a sense that goes beyond self-interested calculations that recognize mutual dependence—especially given recent moves at Nissan and even Toyota toward “less Japanese” supplier relationships.⁴⁶

It is also worth noting that the broader context in which a business operates can create a different type of holdup threat: one reflecting unilateral **expropriation** rather than bargaining based on resource dependence or cospecialization. Social norms about acceptable levels of profit and market power, in particular, can generate legal and other nonmarket challenges that threaten the appropriation of added value. Legal restrictions are worth understanding in some detail here and in the context of some of the discussions in the preceding chapters, because along a number of dimensions, the law can be more constricting than personal conceptions of what is fair or ethical. And governmental

intervention is especially worth watching out for in certain types of industries. These include industries that already feature a high level of governmental involvement (through procurement, funding, regulation, public ownership, etc.), loom large in terms of size, investments, employment, votes, or general political salience (e.g., media ownership), are considered part of the national patrimony (e.g., natural resources), or supply mass consumption staples or “entitlement” goods (e.g., food or even pharmaceuticals). Of course, in such environments, companies often strive to be involved in rule-making. At a broader level, while the origins of nonmarket threats of holdup differ from those of the market-based threats on which this section focuses, some of the same remedies (with the obvious exception of vertical integration) can be employed to mitigate them.

To summarize, holdup is a systematic threat to the appropriability of added value that tends to be based on resource dependence or cospecialization. Strategic responses to holdup vary in the extent to which they emphasize competition versus cooperation. Given the traditional strategic bias toward taking the competitive approach and increasing one’s slice of the pie, explicit attention should also be paid to the possibility of growing the pie. Or to repeat one of the messages from Chapter 2, strategists must think about relationships both cooperatively and competitively.

THE THREAT OF SLACK

Slack is an internal threat to the appropriation of added value that can be defined as the gap between the value actually appropriated over time and the amount potentially available given the other threats to sustainability discussed earlier (and the strategic responses to them). In dynamic terms, slack reflects persistent suboptimization by a business that dissipates appropriable added value or even reduces added value over time. In some contexts, it is more natural to discuss the inverse of slack, what might be referred to as organizational effectiveness.

Although slack is relatively easy to define, it can be hard to measure. Distinguishing between slack and poor choices about how to deal with other threats to sustainability, while desirable, is often difficult. There are, in addition, a host of other complications. Some “slack” may be needed to attract customers (e.g., law offices). Some may be essential for experimentation with new strategies and innovative projects, which might not be possible in a more resource-constrained environment. Some may reflect nonmonetary compensation to employees. And so on. The broader point is that some apparent slack may be of considerable value.

These complications notwithstanding, researchers have tried to measure slack anyway. Statistical estimates, while not very precise, suggest that various inefficiencies dissipate between 10 percent and 40 percent of revenues in U.S. manufacturing.⁴⁷ Large amounts of slack are also suggested by the capability differences among direct competitors that were cited in Exhibit 5.3. But while slack is of general significance, it is of particular interest here because of its relationship to sustained success. Without current or past success, a business is unlikely to have much room to suboptimize (although well-funded start-ups may be exceptions to this rule). Success in the form of appropriable added value expands the scope for suboptimization and can also—as a result of arrogance or other corporate sins—compromise the quality of choice. In plainer language, rich diets are relatively likely to lead to a hardening of corporate arteries. So it is unsurprising that the most stupendous contemporary case

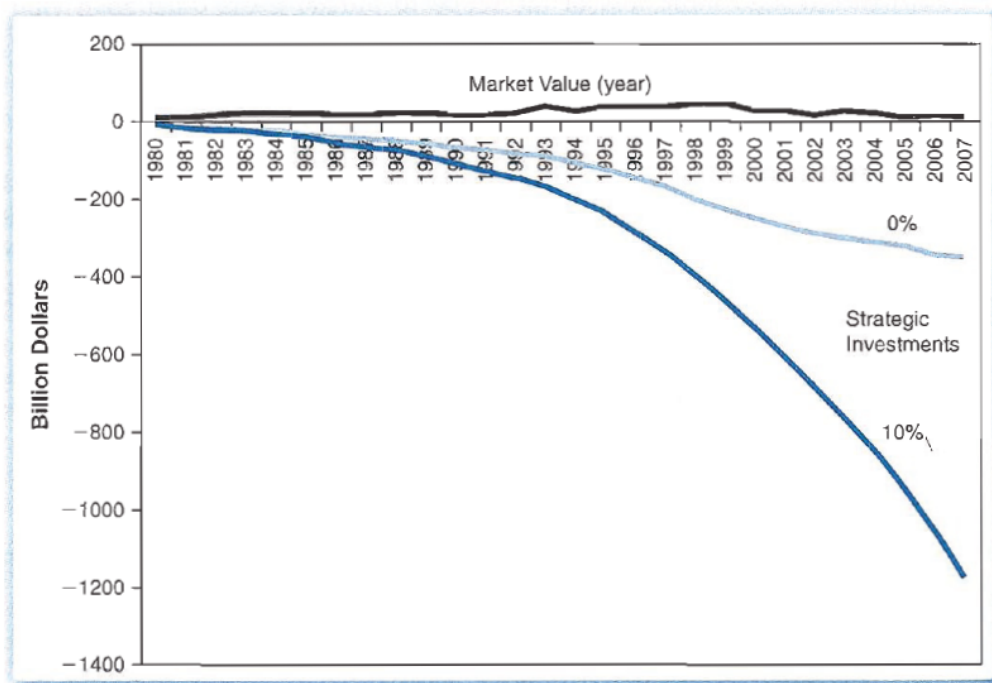
of slack involves a company that was, for decades, the world's largest industrial enterprise, General Motors.

GM's financial performance has been abysmal for some time now. Its manufacturing operations have lost money for the last few years and, at the end of 2007, its equity market value was \$14 billion, barely up from \$13 billion at the end of 1980—and down by more than one-half in real terms. By the end of 2008, that figure had dipped to \$2 billion, and bankruptcy loomed.

But between 1980 and 2007, GM's strategic investments—net capital expenditures plus R&D—amounted to \$346 billion if simply added up and \$1,172 billion in terms of their present value in 2007 if carried forward at a 10 percent annual rate (see Exhibit 5.4)! Even if one assumes that the company's market value would have collapsed to zero in the absence of these strategic expenditures, and that it would have been unable to pay stockholders the \$35 billion in dividends that it did, the amount of value destroyed at GM over this period ran into hundreds of billions of dollars.⁴⁸ In other words the benefit-to-cost ratio of GM's strategic investments between 1980 and 2007 was *much much* lower than one. As H. Ross Perot, briefly a GM director, pointed out, the company could have bought Toyota and Honda in the mid-1980s for *half* of the \$80 billion it spent on a program to automate its factories.

GM's poor performance partly reflects industry effects: average profit margins in the auto assembly industry have declined from 20 percent in the 1930s to around 10 percent in the 1960s and zero to negative levels in the 2000s. But that is not the whole

EXHIBIT 5.4 Market Value versus Cumulated Strategic Investments at General Motors.



story: the Japanese Big Three, particularly Toyota, have performed well financially over the last two decades, and even the smaller of the U.S. Big Three did not destroy as much value as GM in proportionate terms. So one must also look at GM-specific explanations.

There are many such explanations, of which only a few can be cited here. GM failed to respond effectively to the surge in sales of light trucks, and many of its car designs continue to be boxy. Its factory automation program proved both very costly and ineffective. It maintains an oversized production infrastructure in the hope of reversing the erosion of its market share. It continues to receive worse marks for its supplier relationships than Chrysler and Ford. Its relationship with the UAW has arguably been worse as well and has involved greater direct exposure, given a history of more backward integration—although the decline of labor and related expenses, from one-third of revenues in the mid-1980s to one-ninth by 2003, cast some doubt on the company's tendency to emphasize labor holdup as *the* problem.

Whatever the "real" explanation for GM's poor performance, it seems clear that value destruction on this scale would have been impossible if GM had not previously accumulated a pool of resources that afforded it large **free cash flows**. Its ability to destroy several times its assessed market value at the beginning of the 1980s presumably reflects the fact that expectations of high, ongoing levels of slack were also built into the valuation of these cash flows. And GM's ongoing struggles with slack more than a quarter century after Japanese competition shook up the U.S. auto industry suggest that slack is often sustained by powerful inertial forces. As one might expect, GM scores high on most of the indicators of inertia flagged on the basis of organization theory in Exhibit 4.5—size, maturity, complexity, bureaucratization, thick culture, insularity, poor information, and continuity at the top—as well as on governance-related indicators of the scope to pursue goals other than value-maximization. Organization theory also suggests some remedies for sustained slack that will briefly be mentioned here.⁴⁹

Generating Information

The difficulty of measuring slack increases, rather than decreases, the importance of generating information about it. **Benchmarking** against direct competitors or against companies that are best-in-class in certain activities is obviously useful. Simulation, experimentation, and direct investigation of the effects of changes are other ways of learning about opportunities for improvement. But an added emphasis on information gathering is unlikely to be an adequate basis for identifying all such opportunities because of what Oliver Williamson has referred to as **impacted information**: a condition in which one party to a transaction or relationship is much better informed than the other(s).⁵⁰

Monitoring Behavior

A second response to slack, often complementary to the information-gathering option, is to increase the amount of resources devoted to monitoring behavior. Here the goal is to **catch inappropriate behavior before it occurs or to decrease its attractiveness by increasing the probability of detection, backstopped by penalties** (or by rewards for good behavior). One standard example is making workers punch time clocks and docking their pay if they arrive late or quit early. Note, however, that monitoring provides only limited benefits when a wide range of discretionary legitimate choices exist or

when resources are what are referred to as plastic. Is a software developer sitting beside a brook with her eyes closed slacking off or having a commercially valuable vision?

Offering Performance Incentives

Even when monitoring behavior is impractical or uneconomical, it may be possible to reward good behavior indirectly by rewarding good performance. Such an approach works best when an individual's (or group's) behavior is tightly connected to the performance outcomes that are actually observed. This condition is often violated, though, when there are significant coordination requirements or other reasons why performance can be measured only in highly aggregated terms. In addition, a high degree of emphasis on financial incentives can upset norms (especially about equity), exacerbate attempts to game the system, and divert creativity from real to paper entrepreneurialism.

Shaping Norms

A fourth approach to dealing with slack involves supplementing economic rewards or punishments with appeals to **norms, values, a sense of mission**, and so forth. Underlying this approach is the idea that people in organizations are motivated by more than just sticks or carrots. Of course, **moral suasion** alone is unlikely to be totally effective. Given the heated debate between economists and other social scientists about the relative efficacy of economic versus "intrinsic" motivation, the safest conclusion is that a manager intent on reducing slack should consider both types of levers as opposed to fixating on only one of them.

Bonding Resources

Bonding resources is a **top-down approach** to containing slack. It is derived from Michael C. Jensen's theory of the agency costs of free cash flow.⁵¹ According to Jensen, **managers are imperfectly policed by shareholders, have incentives to grow the resources under their control, and are most able to take such steps when free cash flow is large—leading to investments that destroy shareholder value. One obvious remedy is to pile up debt so as to reduce free cash flow** (by creating contractual obligations to pay fixed interest expenses). The risk is that the company may become overloaded with debt (i.e., end up with negative free cash flow).

Changing Governance

Bonding resources is just one of several top-down ways of dealing with slack. Others include creating small but well-informed and powerful boards of directors, restricting the ability of CEOs and other insiders to dominate those boards, requiring board members and top managers to own substantial amounts of a firm's equity (in relation to their personal wealth), encouraging (other) large, active investors, and unwinding cross-subsidies.⁵²

Mobilizing for Change

Forcing change at the top may often be necessary to reverse slack-related problems, but it is rarely sufficient by itself. Research on change management suggests a number of complementary tactics that address inertial barriers: creating dissatisfaction with the status quo, communicating a powerful vision of what can be accomplished with change, and changing people and processes as well as organizational structure.⁵³

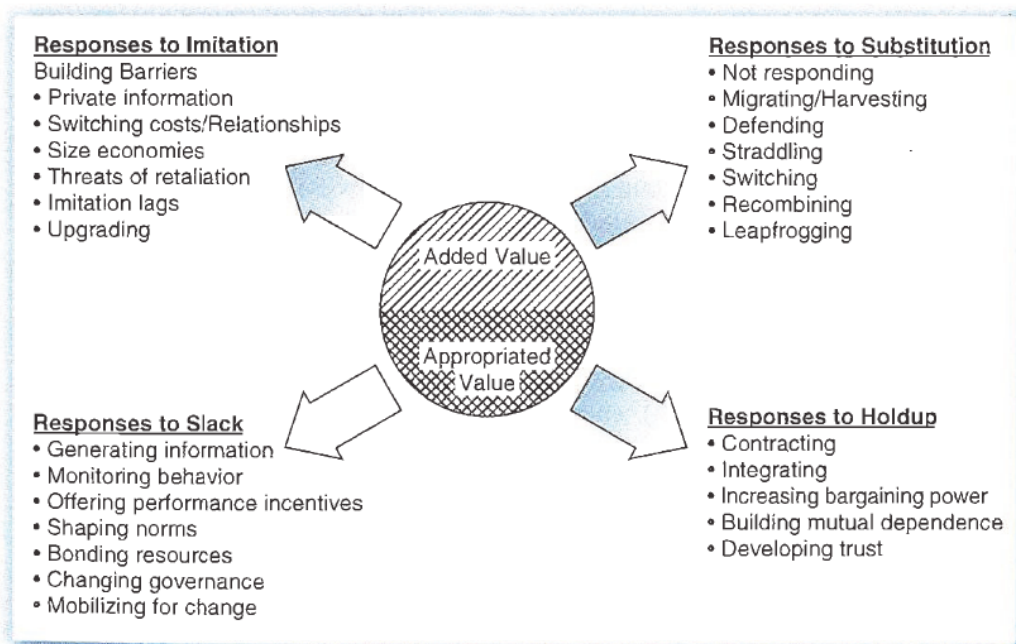
In summary, **slack** is an **internal rather than external threat** to the appropriation of added value. This does not imply, however, that slack is easier to control than the other threats to sustainability discussed in this chapter. The scope for slack is highest in companies that have enjoyed, or are enjoying, considerable economic success. It is amplified by the difficulties of gathering information, offering high-powered incentives, or otherwise directing the organization toward value creation and capture instead of value dissipation.

AN INTEGRATIVE EXAMPLE: WAL-MART

Exhibit 5.5 summarizes the four threats to sustainability as well as responses to them. Note, however, that these threats have so far been discussed in isolation. An integrated perspective on them is of interest, since companies typically face more than one threat at a time. For an integrative example that also illustrates the value of systematically thinking through threats to sustainability even in a situation where the scope for sustainability might seem relatively limited, consider the case of Wal-Mart, which posted a net income of \$11.3 billion on sales of \$345 billion in 2007, and had a market value at the end of the year of \$190 billion. These numbers made Wal-Mart the world's largest company and, with 1.9 million employees, the world's largest private employer as well.

To begin by linking back to Chapter 3, Wal-Mart's competitive strategy has been to outperform its competitors by achieving lower operating costs and higher asset turns/efficiency (i.e., lower fixed costs per unit). The emphasis on low costs—and the

EXHIBIT 5.5 Responding to Threats to Sustainability



associated themes of buying in bulk, pricing low, focusing on metrics such as sales per square foot, and operating frugally, among others—have been in place at Wal-Mart for a long time now. In fact, these themes predate the startup of the first Wal-Mart store in 1962; they also characterized the Ben Franklin stores that Wal-Mart's founder, Sam Walton, operated as a franchisee in the 1940s and 1950s. So it seems to make sense to think of Wal-Mart as having been designed around a broad strategic target, low cost, with some continuity in terms of its definition and pursuit, but also some evolution in the context of an adaptive, analytical culture that searches out and spreads improvements across many physical locations.

Given Wal-Mart's focus on *discount* retailing, it is not very surprising that a low-cost/low-price position has proven to be the biggest engine of economic value creation within that industry. What *is* more surprising is Wal-Mart's ability to sustain superior performance with this apparently obvious strategy: it has posted a return on equity in excess of 20 percent in all but a handful of years since the early 1970s while growing revenues a thousand-fold. Why *hasn't* this successful strategy fallen prey to the four threats to sustainability: imitation, substitution, holdup, and slack?

To begin with imitation, Wal-Mart is protected by an array of barriers, some of which it has deliberately erected:

- **Scale Economies:** Wal-Mart initially tapped local scale economies by locating in towns large enough to accommodate one discounter but not two. It moved on to exploit regional scale economies by building up its hub-and-spoke distribution system, followed by the pursuit of national scale economies in dealing with suppliers and intermediaries.
- **Learning/Private Information:** Wal-Mart has gone to great lengths to analyze the myriad details of retailing: how to load trucks to minimize storage space, unpack boxes, schedule part-time labor, juggle fixture heights for maximum impact, and so on. In addition, it aggressively "borrows" ideas from all over and maintains the second-largest customer database in the world.
- **Switching Costs/Relationships:** Wal-Mart is often perceived as having the lowest prices even in markets where it doesn't. Lock-in is also increased by its ability to persuade suppliers to make major Wal-Mart-specific investments in systems such as Retail Link, which furnishes them with computerized access to real-time data on the sales and inventories of their products at the store level.
- **Threats of Retaliation:** Wal-Mart's policy of allowing store managers to match or beat the lowest competing price on an item in their trading area by as much as 5 percent has presumably had a chilling effect on competitors/potential entrants in its local markets, as has its general reputation for fierceness.
- **Lags/Upgrading/Complexity:** Wal-Mart has kept competitors off balance by continuously upgrading its capabilities, particularly in information technology and logistics, and making discrete resource commitments to initiatives such as Retail Link. Imitating Wal-Mart's culture or its entire complex activity system would probably take a very long time even if it *were* feasible.

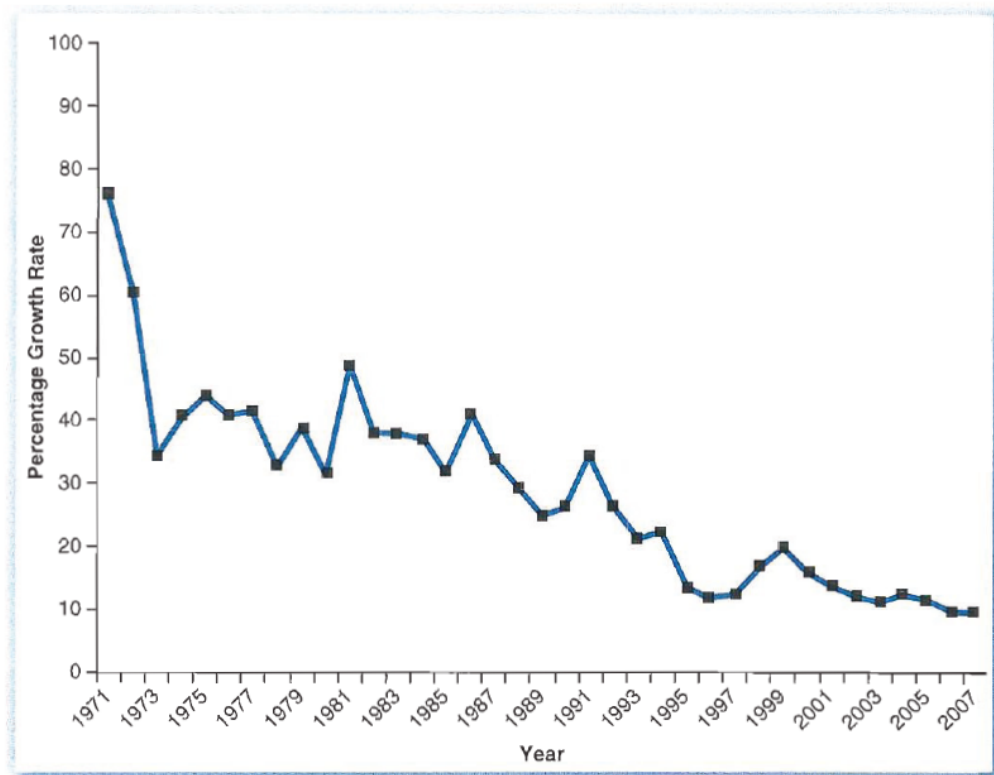
The principal substitution threats that Wal-Mart has faced have been related to the risk that its discount retailing operations might be replaced by a new format. This kind of format risk is highlighted by the theory of the "wheel of retailing," which predicts that operators of old formats will have trouble adjusting to the new ones. But Wal-Mart

has actually been quite active in exploring potential substitutes for its traditional discount stores: it entered warehouse clubs in a big way, enjoyed enormous success with the hybrid food-nonfood format of Supercenters, and is exploring a smaller format with Neighborhood Markets. This kind of mobility across formats should not be too surprising, because Sam Walton started up Wal-Mart as a way of switching from one retailing format to another. Wal-Mart also tries to be a leader in figuring out how to deploy new technologies with transformational potential—such as RFID or wireless tags that can be used to track goods passively as they pass from suppliers to warehouses to stores.

As for holdup, it is useful to distinguish between suppliers of merchandise and employees. In regard to the former, Wal-Mart's extraordinary growth has allowed it to build up an unusual amount of bargaining power, even vis-à-vis very large suppliers. Thus, in 2007, Procter & Gamble derived 15 percent of its total revenues from sales to Wal-Mart, while the same dollar amount represented 3.3 percent of Wal-Mart's revenues.⁵⁴ Wal-Mart does bargain hard with its suppliers by, for example, focusing on a single pricing number; getting suppliers to pay for modular displays, promotions, and the like, on top of that; and developing private labels after learning the business from them. But it also seems to afford suppliers enough surplus—in terms of real-time information and shared expertise as well as volume and terms—to induce them, for example, to invest much more in the aggregate in its Retail Link platform than the \$4 billion that Wal-Mart itself reportedly spent. And companies that supply Wal-Mart seem to perform better than those that focus on other retailers.⁵⁵

Labor holdup is more of an issue. Wal-Mart has fiercely and so far successfully resisted unionization efforts. Holding the line on wages is critically important, since a \$1 per hour across-the-board increase might reduce Wal-Mart's income by one-quarter. But there has been enormous negative fallout, ranging from employee demotivation and very high turnover rates to unflattering public portrayals—for example, of Wal-Mart employees living below the poverty line or lacking health-care coverage. Behind the public portrayals is heightened public scrutiny of Wal-Mart now that it has become the largest company in the world and a related shift in public perceptions of the company—in former CEO David Glass's words—from endearing underdog to overbearing top dog. Wal-Mart has moved to address these issues and there are indications of some recent success in this regard. That said, public perceptions of Wal-Mart's penny-pinching and power are likely to continue to be concerns—and possible catalysts of nonmarket holdup—for the foreseeable future.

Finally, Wal-Mart seems to have experienced limited slack so far, for a number of reasons. The scope for deliberate departures from value-maximization is limited by the Walton family, which owns a little over 40 percent of the company's stock and whose interests are represented by Sam Walton's eldest son, Rob Walton, who serves as chairman of Wal-Mart's board of directors. Employees are motivated through incentives; particularly at the store-manager level and above, tight performance monitoring that is facilitated by the pursuit of a cost-based strategy, and social control through deeply embedded norms about frugality, performance improvement, and so on. Nonetheless, slack may become a significant issue in the future: because of the challenges of communicating with two million employees well enough to preserve at least elements of Wal-Mart's traditional culture, doing so despite weakening bonds between store managers and employees because of larger stores and 24-hour operation, dealing with widening horizontal spans of control and increasing bureaucratic complexity (especially given the centralization of decision making at corporate headquarters), and similar factors.

EXHIBIT 5.6 Wal-Mart Revenue Growth, 1971–2007.

In addition to creating organizational challenges for Wal-Mart, growth has become a challenge in its own right (see Exhibit 5.6). Wal-Mart's declining growth rate reflects its increasingly large share of a retailing category, discounting, that has grown at only 2–3 percent per year recently. The reason Wal-Mart has done as well as it has in the last decade is because it has built up a major growth engine outside its traditional core with its Supercenter format. It now faces the challenge of developing another new growth engine or two by accelerating expansion into other businesses or geographies. Such decisions about horizontal and geographic scope are, respectively, the topics of Chapters 6 and 7.

Summary

Like Chapter 4, this chapter concentrated on dynamics. Unlike Chapter 4, it focused on situations in which players are more numerous or faceless and identified four evolutionary dynamics that threaten sustainability in such contexts. Two dynamics—imitation and substitution—threaten businesses' added value, and two others—holdup and slack—threaten

their owners' ability to appropriate that added value for themselves. Understanding these dynamics and looking across all of them should afford managers extra insight into the evolution of the business landscapes on which they operate and the issue of how to counter threats—or even turn them into opportunities.

Glossary

appropriability	expropriation	organizational
benchmarking	free cash flow	effectiveness
capability	holdup	retaliation
causal ambiguity	imitation	scarcity value
complexity	impacted	slack
commitment	information	social complexity
contestable	interorganizational	sticky resources
cospécialization	relationships	substitution
cross-sectional fit	learning	sustainability
disruptive technologies	moral suasion	tetra-threat framework

Notes

1. These data were originally reported in Pankaj Ghemawat, *Commitment: The Dynamic of Strategy* (New York: Free Press, 1991).
2. Robert R. Wiggins and Timothy W. Ruefli, "Sustained Competitive Advantage: Temporal Dynamics and the Incidence and Persistence of Superior Economic Performance," *Organization Science* 13, no. 1 (January–February 2002): 82–105.
3. Anita M. McGahan and Michael E. Porter, "The Emergence and Sustainability of Abnormal Profits," *Strategic Organization* 1, no. 1 (February 2003): 79–108.
4. Ghemawat, *Commitment*, chap. 2.
5. The basic idea can be found in Kenneth R. Andrews, *The Concept of Corporate Strategy* (Homewood, Ill.: Richard D. Irwin, 1971), but was revived in 1984 in an article by Birger Wernerfelt, "A Resource-Based View of the Firm," *Strategic Management Journal* 5, no. 2 (April–June 1984): 171–180.
6. For example, Wernerfelt, "Resource-Based View," defined resources very broadly as "anything which could be thought of as a strength or weakness of a given firm" (p. 172).
7. Consider, for example, what seems to be the most popular resource-based framework for assessing sustained strengths and weaknesses, the VRIO framework proposed by Jay Barney, where V denotes value, R rarity, I inimitability, and O organizational effectiveness. V, R, and I seem to have a high degree of overlap, especially in a dynamic context, although this framework *does* provide some interesting ways of thinking about I and about complexity that are briefly discussed later in this chapter, in the section on the threat of imitation. See Jay B. Barney, *Gaining and Sustaining Competitive Advantage*, 2nd ed. (Upper Saddle River, N.J.: Prentice Hall, 2002), chap. 5, pp. 149–192.
8. Steven A. Lippman and Richard P. Rumelt, "A Bargaining Perspective on Resource Advantage," *Strategic Management Journal* 24, no. 11 (November 2003): 1069–1086.
9. To make the connections with Chapter 4 precise, this is equivalent to replacing game-theoretic equilibrium in strategies with equilibrium defined in terms of observed outcomes and underpinned by the behavioral assumption of "one smart rival." The latter assumption embodies much weaker rationality requirements than Nash equilibrium, as explained in John Sutton, "One Smart Agent," *Rand Journal of Economics* 28, no. 4 (Winter 1997): 605–628.
10. Adam M. Brandenburger and Barry J. Nalebuff, *Co-Opetition* (New York: Doubleday, 1996), chaps. 5 and 6.
11. The evidence on imitation is discussed in more detail in Pankaj Ghemawat, "Sustainable Advantage," *Harvard Business Review* 64, no. 5 (September–October 1986): 53–58. The article provides specific citations.
12. Stephen P. Bradley, John R. Wells, and James Weber, "The Pharmaceutical Industry in 2003," (Harvard Business School Publishing Case no. 704-470, Boston, 2004).
13. Leila Abboud, "Raging Hormones: How Drug Giant Keeps a Monopoly on 60-Year-Old Pill," *Wall Street Journal*, September 9, 2004, p. A1.
14. The details in this paragraph are based on Marcia Angell, *The Truth About the Drug Companies: How They Deceive Us and What to Do About It* (New York: Random House, 2004).
15. It is also worth noting the related category of *network externalities*, which exist when the

- attractions to buyers, suppliers, or complementors of joining a network increase with its size.
16. For an example of the pressure that this has put on large generic manufacturers, see the discussion of Teva of Israel in Leila Abboud, "Clone Wars: An Israeli Giant in Generic Drugs Faces New Rivals," *Wall Street Journal*, October 28, 2004, p. A1.
 17. Citations for most of these sources can be found in Ghemawat, *Commitment*, chap. 5. See also Richard Hall, "The Strategic Analysis of Intangible Resources," *Strategic Management Journal* 13, no. 2 (February 1992): 135–144.
 18. Public Citizen, "Rx R&D Myths: The Case against The Drug Industry's R&D 'Scare Card'" (2001 [accessed November 2, 2004]); available from www.citizen.org/documents/ACFDC.PDF, cited in Angell, *Truth About the Drug Companies*, p. 269, fn. 7.
 19. Jeffrey R. Williams, "How Sustainable Is Your Competitive Advantage?" *California Management Review* 34, no. 3 (Spring 1992): 29–51.
 20. Steven A. Lippman and Richard P. Rumelt, "Uncertain Imitability: An Analysis of Interfirm Differences in Efficiency under Competition," *Bell Journal of Economics* 13, no. 2 (Autumn 1982): 418–438.
 21. Jay Barney, "Firm Resources and Sustained Competitive Advantage," *Journal of Management* 17, no. 1 (March 1991): 99–120.
 22. Michael E. Porter, "What Is Strategy?" *Harvard Business Review* 74, no. 6 (November–December 1996): 61–78.
 23. W. Carl Kester, "Today's Options for Tomorrow's Growth," *Harvard Business Review* (March–April 1984): 153–160.
 24. For a book-length treatment of commitment decisions that also discusses how uncertainty can be folded into the analysis—basically, through the concept of flexibility value—see Ghemawat, *Commitment: The Dynamic of Strategy* (New York: Free Press, 1991), esp. chap. 6.
 25. For the original discussion that frames these and other issues concerning capabilities, see David J. Teece and Gary P. Pisano, "The Dynamic Capabilities of Firms: An Introduction," *Industrial and Corporate Change* (1994 No. 3), pp. 537–556.
 26. See, for example David M. Upton, "Flexibility as Process Mobility: The Management of Plant Capabilities for Quick Response Manufacturing," *Journal of Operations Management* 12, nos. 3–4 (June 1995): 205–224; Rebecca Henderson and Iain Cockburn, "Scale, Scope, and Spillovers: The Determinants of Research Productivity in Drug Discovery," *Rand Journal of Economics* 27, no. 1 (Spring 1996): 32–59; Kim B. Clark and Takahiro Fujimoto, *Product Development Performance: Strategy, Organization, and Management in the World Auto Industry* (Boston: Harvard Business School Press, 1991), and Gary P. Pisano, *The Development Factory: Unlocking the Potential of Process Innovation* (Boston: Harvard Business School Press, 1997).
 27. For some tips on how to conduct a capability audit, see Dave Ulrich and Norm Smallwood, "Capitalizing on Capabilities," *Harvard Business Review* 82, no. 6 (June 2004): 119–126.
 28. See Dorothy Leonard-Barton, "Core Capabilities and Core Rigidities: A Paradox in Managing New Product Development," *Strategic Management Journal* (1992): 111–125.
 29. Joseph A. Schumpeter, *Capitalism, Socialism, and Democracy* (New York: Harper, 1942), p. 84. See also Ingemar Dierickx and Karel Cool, "Asset Stock Accumulation and Sustainability of Competitive Advantage," *Management Science* 35, no. 12 (December 1989): 1504–1511.
 30. Clayton M. Christensen, *Innovation and the General Manager* (Boston: Irwin/McGraw-Hill, 1999), idem, *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail, The Management of Innovation and Change Series* (Boston: Harvard Business School Press, 1997).
 31. Christensen, *Innovator's Dilemma*.
 32. Ibid.
 33. Richard N. Foster, *Innovation: The Attacker's Advantage* (New York: Summit Books, 1986).
 34. W. Chan Kim and Renee Mauborgne, "Value Innovation: The Strategic Logic of High Growth," *Harvard Business Review* 75, no. 1 (January–February 1997): 102, reprinted in June 2004 issue.
 35. This comparison and the preceding information on the Ambassador Bridge are based on John Lippert and Erik Schatzker's "Matty's Bridge," Bloomberg, March 2003.
 36. "Michigan Gives Thumbs Up to Twin Ambassador Span," Today'sTrucking.com, March 16, 2007, at <http://www.todaystrucking.com/news.cfm?intDocID=17626>; "Ambassador Bridge Representatives Lash Out," June 13, 2008, at <http://liberalmedianot.blogspot.com/2008/06/ambassador-bridge-representatives-lash.html>; "Ambassador Bridge To Expand," *Detroit News*, June 17, 2008, at <http://www.clickondetroit.com/news/16631725/detail.html>;
 37. For the pioneering discussion of holdup, see Oliver E. Williamson, *Markets and Hierarchies, Analysis and Antitrust Implications: A Study in the Economics of Internal Organization* (New York: Free Press, 1975).

38. For a historical overview on which some of this discussion is based, see Susan Helper, "Strategy and Irreversibility in Supplier Relations: The Case of the U.S. Automobile Industry," *Business History Review* 65, no. 4 (1991): 781–824.
39. Reasons include bounded rationality, uncertainty about the future, and asymmetric information. See Williamson, *Markets and Hierarchies*, and the discussion in Chapter 6.
40. William J. Abernathy and Kenneth Wayne, "Limits of the Learning Curve," *Harvard Business Review* 52, no. 5 (September–October 1974): 109–119.
41. Michael E. Porter, "Note on Supplying the Automobile Industry (Condensed)" (Harvard Business School Publishing case no. 386-176, Boston, 1986).
42. Jeffrey H. Dyer, "Does Governance Matter? Keiretsu Alliances and Asset Specificity as Sources of Japanese Competitive Advantage," *Organization Science* 7, no. 6 (November–December 1996): 649–666.
43. Ibid.
44. For a general discussion of how interorganizational relationships can lead in this fashion to sustained competitive advantage, see Jeffrey H. Dyer and Harbir Singh, "The Relational View: Cooperative Strategy and Sources of Interorganizational Competitive Advantage," *Academy of Management Review* 23, no. 4 (October 1998): 660–679.
45. Michelene Maynard, "Auto Parts du Jour; Suppliers Are Forced to Learn Flexibility," *New York Times*, August 31, 2001; and Andreas Maurer, Frank Dietz, and Nikolaus S. Lang, "Beyond Cost Reduction: Reinventing the Automotive OEM-Supplier Interface" (Boston Consulting Group, 2004): 18.
46. See, for instance, Pankaj Ghemawat and Rajiv Shukla, "The Coevolution of Auto Supply Chains: The U.S. and Japan" (Harvard Business School Publishing case no. 700-062, Boston, 2000).
47. Consult, for example, Richard E. Caves and David R. Barton, *Efficiency in U.S. Manufacturing Industries* (Cambridge, Mass.: MIT Press, 1990).
48. For more details on this methodology and an application to General Motors between 1980 and 1990, see Michael C. Jensen, "The Modern Industrial Revolution, Exit, and the Failure of Internal Control Systems," *Journal of Finance* 48, no. 3 (1993): 831–880.
49. For a more extended discussion of some of the ideas touched on here, see Paul R. Milgrom and John Roberts, *Economics, Organization and Management* (Englewood Cliffs, N.J.: Prentice-Hall, 1992), chap. 6.
50. Williamson, *Markets and Hierarchies*, 14.
51. Michael C. Jensen, "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers," *American Economic Review* 76, no. 2 (1986): 323–329.
52. Jensen, "Modern Industrial Revolution, Exit, and the Failure of Internal Control Systems."
53. For a managerial discussion, see John P. Kotter, "Leading Change: Why Transformation Efforts Fail," *Harvard Business Review* 73, no. 2 (March–April 1995): 59–67.
54. Proctor & Gamble's acquisition of Gillette did not change their situation much: Wal-Mart accounted for 16-1 of P&G and Gillette's combined revenues in 2004, and their sales to it represented 14-1 of Wal-Mart's revenues.
55. Marco Iansiti and Roy Levien, *The Keystone Advantage: What the New Dynamics of Business Ecosystems Mean for Strategy, Innovation, and Sustainability*. (Boston: Harvard Business School Press, 2004).